

Searching for Tau Core signal candidates in SPARC residual structure

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Abstract

Paper 1 found that externally reviewed structural disturbance is associated with larger low-acceleration residual scatter in SPARC. Paper 2 reversed the question and showed that fixed residual-shape features can recover those A/C labels better than chance, while remaining explicitly non-unique with respect to MOND-simple and empirical RAR-like baselines. This Paper 3 seed asks a narrower follow-up question: where do TPG/projection residuals, MOND-simple residuals, RAR-like residuals, and measured rotation curves point to candidate residual structure that could plausibly carry the missing observer- and environment-dependent Tau Core weight? The current packet identifies candidate galaxies, radial onset categories, and environment/observability stress channels. It does not validate Tau Core, does not numerically test RMOND, and does not claim gravity-model selection.

1 Purpose and claim boundary

The working hypothesis is that the TPG/projection baseline already carries part of the local Tau Core weighting, while the remaining TPG residual may carry missing observer- and environment-dependent weights. This is a candidate-search statement, not a proof. A defensible Paper 3 must therefore separate three layers:

1. an operational residual layer, based on frozen TPG/projection, MOND-simple, and RAR-like residual maps;
2. a systematics layer, based on distance, inclination, point count, HI kinematics, beam smearing, and non-circular motions;
3. a theory layer, where a Tau Core interpretation is allowed only if the residual pattern survives the systematics layer and is more specific than ordinary RAR/MOND behavior.

The permitted claim is that this packet defines reproducible Tau Core signal candidates. The forbidden claim is that the candidates already prove Tau Core.

2 Data inherited from Papers 1 and 2

The packet inherits derived SPARC residual artifacts from the Paper 1 and Paper 2 public packages. Raw SPARC rotmod files and raw HI survey products are not redistributed. The retained derived tables include pointwise absolute residuals for TPG/projection, MOND-simple, and empirical RAR-like families; galaxy-level residual features; distance, observability, and environment summaries; and small radial-pilot control summaries.

The main residual quantity is

$$r_{m,gi} = |\log V_{\text{obs},gi} - \log V_{m,gi}|,$$

with galaxy-level burden

$$\text{RMS}_{m,g} = \sqrt{\frac{1}{N_g} \sum_i r_{m,gi}^2}.$$

The TPG-specific excess used for candidate triage is

$$\Delta_{\text{TPG-low},g} = \text{RMS}_{\text{TPG},g} - \min(\text{RMS}_{\text{MOND},g}, \text{RMS}_{\text{RAR},g}).$$

Positive values mark galaxies where the TPG/projection residual is larger than the best available low-acceleration comparator in this packet. Negative values mark TPG success controls.

3 Theory motivation

The relevant external baseline is the SPARC radial acceleration relation [4, 2] and the broader MOND/RAR literature [5, 3]. These works are not optional background; they define the strongest ordinary low-acceleration competitor. If Tau Core only reproduces the same residual ordering as RAR/MOND, it is not yet distinguished.

The systematics literature is equally central. Non-circular motions in HI velocity fields [8, 7] and beam-smearing/rotation-curve quality effects [1] can generate residual structure without new physics. The JWST/NIRCam Zone-of-Avoidance result [6] is used only as an observer/line-of-sight motivation: foreground obscuration and hidden structure can materially affect what an observer can map. It is not evidence for Tau Core by itself.

4 Candidate construction

For each galaxy we record TPG/projection RMS, MOND-simple RMS, RAR-like RMS, TPG-specific excess, environment proxy, distance, reconstruction-risk proxy, inclination, and the first radial bin where the TPG/projection absolute residual exceeds 0.15 dex. Candidate classes are assigned by frozen screening rules:

- *clean tau candidate*: high TPG residual, high TPG-specific excess, and non-high reconstruction risk;
- *environment tau candidate*: high residual burden with high environment cue;
- *TPG divergence follow-up*: positive TPG-specific excess without enough cleanliness for the first two labels;
- *TPG success control*: low residual burden where TPG is no worse than MOND/RAR-like baselines.

These classes are triage labels. They are not physical classifications.

Table 1: Candidate shortlist from the regenerated seed packet.

Galaxy	Class	TPG RMS	TPG excess	Env.	Onset	Candidate class
UGC00731	C	0.340	0.093	0.00	inner_R<0.33Rmax	clean tau
CamB	A	0.964	-0.080	1.72	inner_R<0.33Rmax	environment
UGC05764	A	0.329	0.076	0.00	inner_R<0.33Rmax	TPG divergence
NGC2403	A	0.085	0.029	1.86	outer_R>=0.67Rmax	TPG divergence
NGC3741	C	0.204	0.103	-0.12	inner_R<0.33Rmax	TPG divergence
UGC08490	C	0.186	0.055	0.00	inner_R<0.33Rmax	TPG divergence
NGC3109	C	0.176	0.039	0.20	middle_0.33-0.67Rmax	TPG divergence
UGC06446	C	0.194	0.038	0.00	inner_R<0.33Rmax	TPG divergence

5 Residual and environment readouts

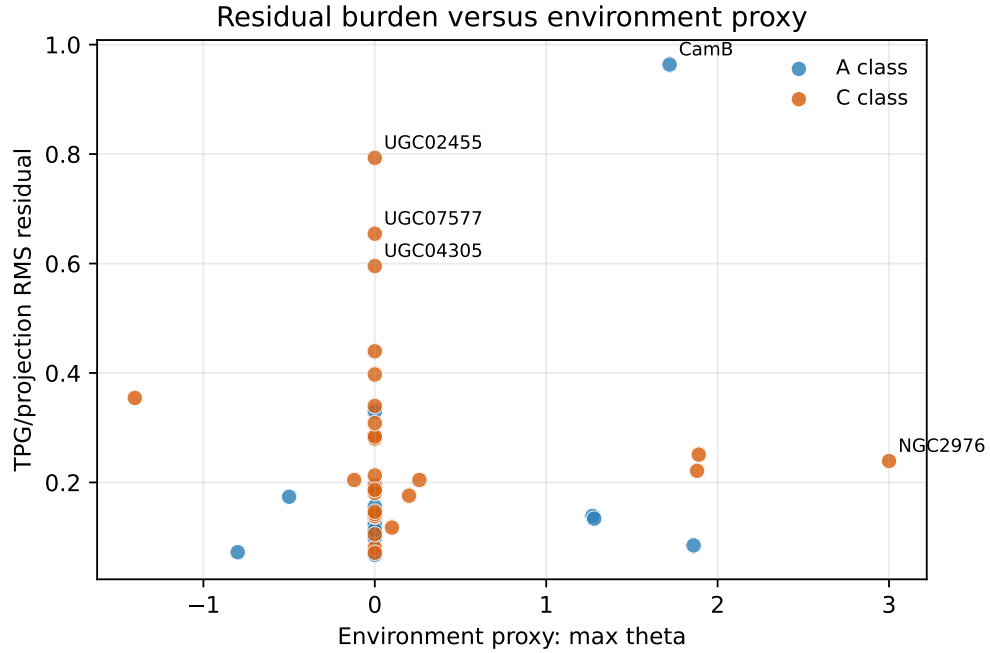


Figure 1: TPG/projection residual burden versus the inherited environment proxy. Named points are the largest seed-candidate scores.

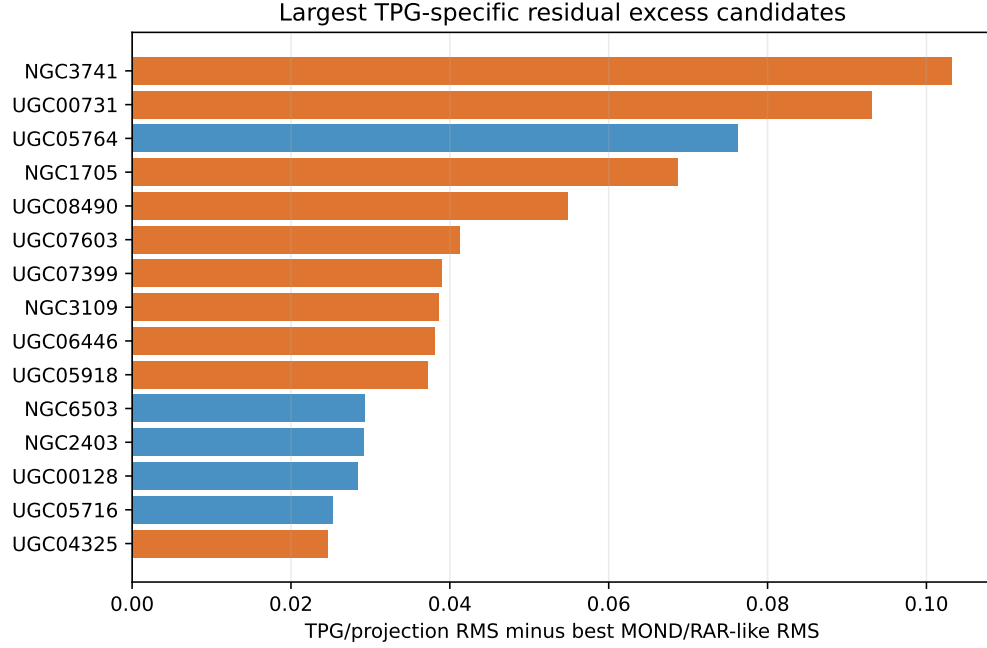


Figure 2: Galaxies with the largest TPG/projection residual excess relative to the better of MOND-simple and RAR-like RMS.

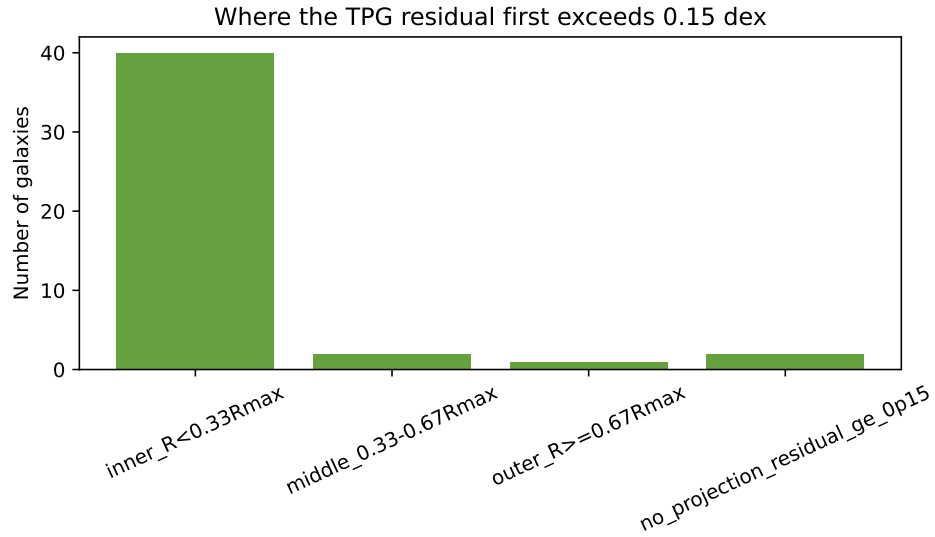


Figure 3: Radial bin where the TPG/projection absolute residual first exceeds 0.15 dex. This is a morphology-of-failure diagnostic, not a detection statistic.

Table 2: Screening correlations for TPG/projection residual burden.

Covariate	Pearson r	Interpretation
DistanceMpc	-0.291	screening_correlation_not_causal_attribution
EnvMaxTheta	0.110	screening_correlation_not_causal_attribution
W_tau_eff_abs_v01	0.983	screening_correlation_not_causal_attribution
ReconstructionRiskChannel_v01	0.317	screening_correlation_not_causal_attribution
MeanErrVobsKms	-0.214	screening_correlation_not_causal_attribution
InclinationErrorDeg	0.214	screening_correlation_not_causal_attribution

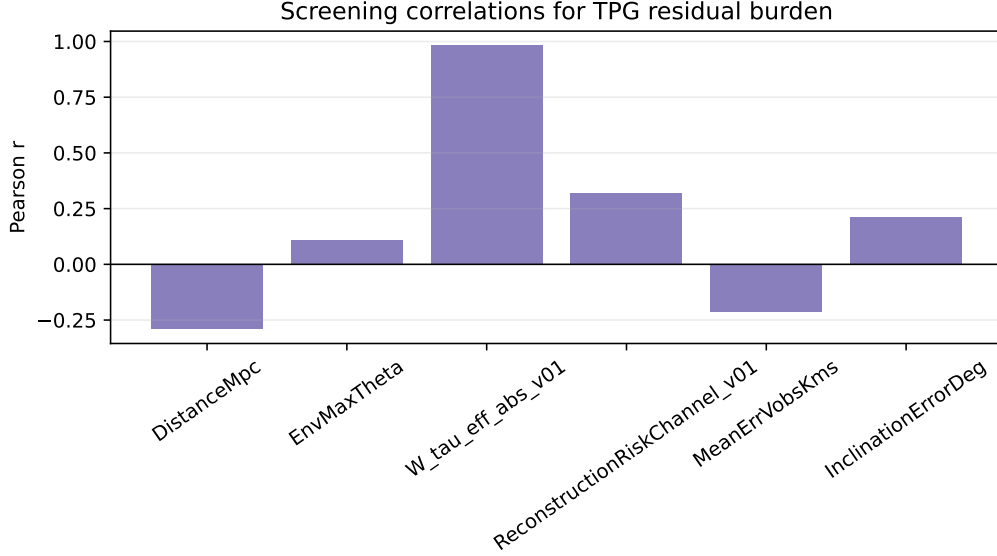


Figure 4: Screening correlations with the TPG/projection residual burden. These are not causal estimates.

6 Required S_τ diagnostic

The extended Tau Core form can be written operationally as

$$F_\tau(a_N, R) = 1 + S_\tau(R) \alpha \ln \left(1 + \frac{a_0}{a_N} \right).$$

For each SPARC point with usable baryonic baseline, the required local value is

$$S_{\tau, \text{req}}(R) = \frac{V_{\text{obs}}(R)/V_N(R) - 1}{\alpha \ln(1 + a_0/a_N(R))}.$$

This is an inverse diagnostic. It uses the measured endpoint and therefore cannot be used as a predictive Tau Core validation. Its role is to ask what kind of future predictive rule would be needed: a galaxy-level constant, a radial function, an acceleration function, or an environment-coupled function.

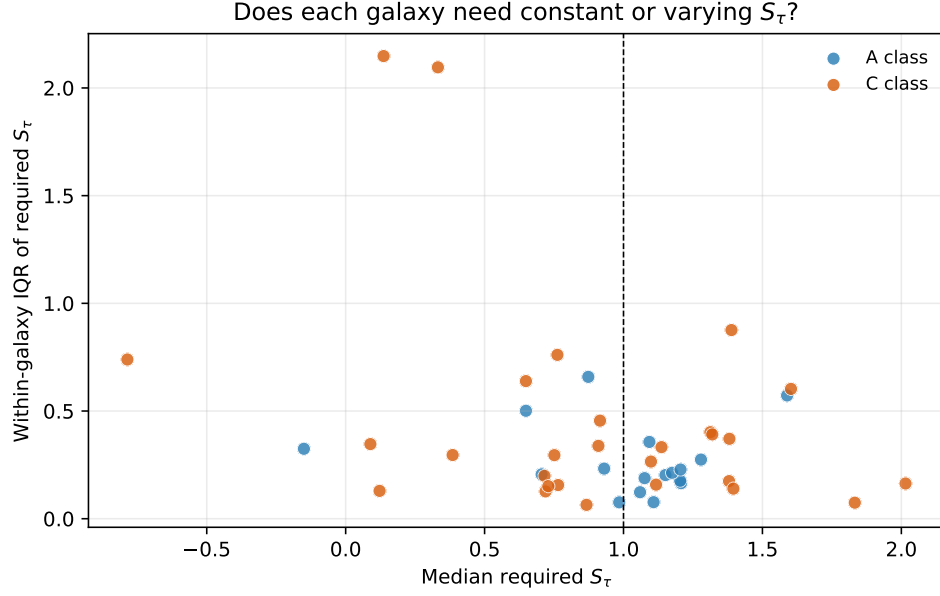


Figure 5: Median required S_τ versus within-galaxy spread. A low spread near $S_\tau = 1$ would support a constant correction; large spread indicates that a function is needed.

Table 3: In-sample required- S_τ function-family diagnostics. Lower RMS means that family can absorb the measured TPG residual more compactly.

Family	Median RMS	Mean RMS	Galaxies
radius + acceleration	0.143	0.152	45
quadratic radius	0.145	0.153	45
linear radius	0.146	0.168	45
quadratic acceleration	0.165	0.160	45
linear acceleration	0.165	0.167	45
galaxy constant	0.187	0.183	45
fixed S=1	0.188	0.249	45

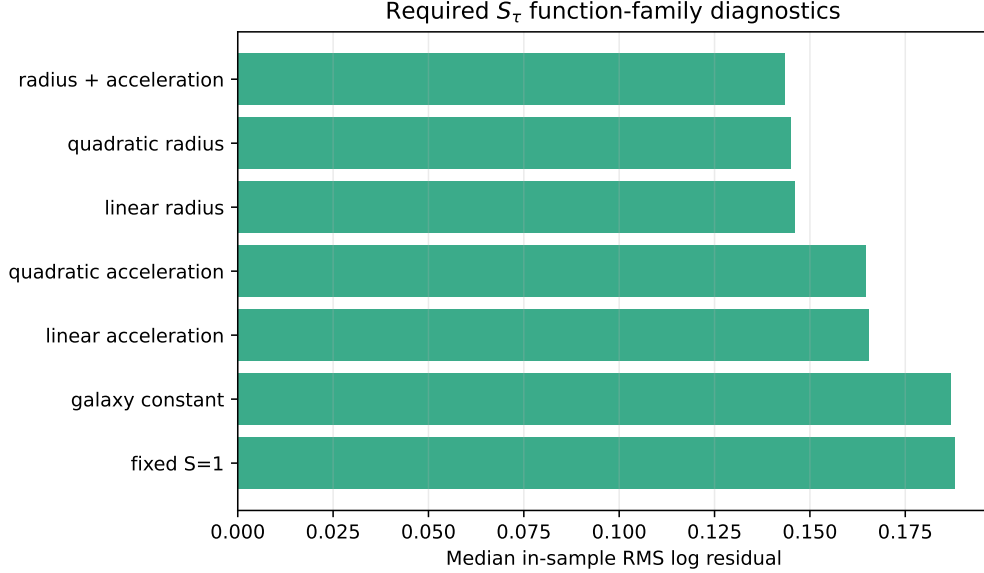


Figure 6: Diagnostic comparison of constant and simple functional S_τ families. This is not model selection because the measured velocities are used to infer S_τ .

7 RMOND and comparator status

The requested RMOND comparison is not yet a numeric endpoint in this seed packet. The local TPG–RMOND bridge is useful, but it is not the same object as a frozen pointwise velocity law. Its strongest result is structural compatibility: the tau-projection can preserve the leading SZ20 scalar and vector Lagrangian exponents under a channel-normalised kernel. That is theory motivation, not a direct prediction for $V(R)$.

The local RMOND–MTW hybrid analysis also warns against double counting. At $f_{\text{vec}} = 0$, the hybrid reproduces pure MTW/TPG exactly, so it is not an independent comparator. At nonzero f_{vec} , the vector contribution requires a frozen halo/vector prescription and screening rule before any residual table is meaningful. Treating either option as a completed RMOND residual test would weaken the paper.

The next technical gate is therefore sharper than “find a CSV”: freeze a unique $V_{\text{RMOND}}(R)$ prescription, regenerate a derived pointwise residual table on the same SPARC radii, and only then add RMOND beside TPG/projection, MOND-simple, and RAR-like baselines.

8 Interpretation

The most useful interpretation is modest. TPG success controls show where the local projection baseline is already adequate. TPG divergence follow-ups show where the residual has structure that might encode missing observer/environment weights, but these are exactly the cases where ordinary systematics can also enter. A candidate becomes interesting only when three facts hold together: the TPG residual diverges in a structured radial way, the divergence is not shared by all low-acceleration baselines, and the object has a plausible Tau Core weighting channel that is not reducible to distance, inclination, beam smearing, or non-circular motion.

9 Conclusion

This seed opens Paper 3 as a reproducible candidate-search project. It carries forward the discipline learned from Papers 1 and 2: freeze the endpoint, compare against MOND/RAR baselines, name the systematics, and avoid promoting a diagnostic pattern into a physical proof. The current evidence is enough to define targets for Tau Core follow-up, especially TPG-divergence galaxies and TPG-success controls, but it is not enough to claim a detected Tau Core field.

The next paper-grade step is a frozen RMOND residual table plus a held-out environment/line-of-sight validation rule. Only after that can the paper ask whether the residual candidates are truly Tau Core-specific rather than ordinary low-acceleration or HI-systematics behavior.

References

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